

Problem overview:

Gotham has experienced a recent surge in crime rates. The mayor and the police want to obtain analysis of crime rates and evidence-based insights to help reduce crime without relying on Batman and his associates. Using the gotham-crime-rate dataset, a time series analysis will be performed, including identification of different patterns, application of transformations and comparison of ARIMA models.

Question 1:

In order for the data to be considered stationary, the following must be true:

$E(X_t) = \mu \rightarrow$ The mean should be constant

$Var(X_t) = \sigma^2 \rightarrow$ Variance should be constant and finite

$Cov(X_t, X_{t+h}) = \gamma(h) \rightarrow$ Covariance must only depend on the lag and not the time

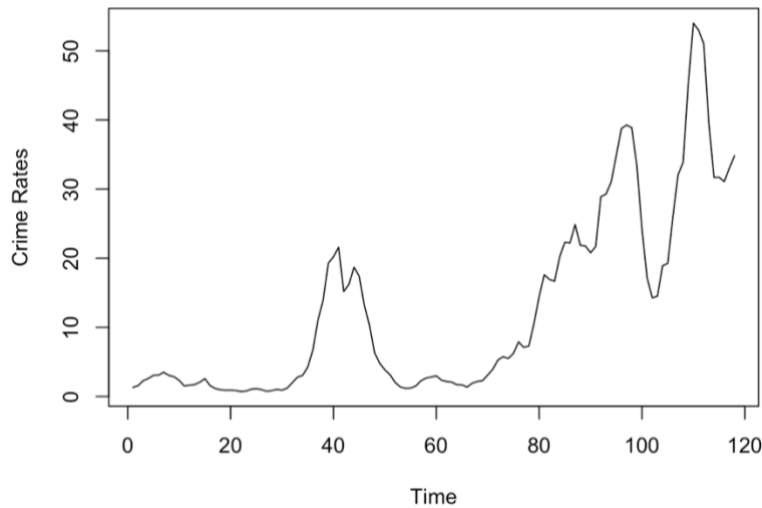


Figure 1: Time Series Plot of Crime Rates vs. Time

Based on Figure 1, there appears to be heteroskedasticity present since the variance fluctuates over time. The mean of the data is not constant and the crime rates seem to have an upward trend with multiple peaks which may suggest seasonal recurrence, therefore the data appears to be non stationary.

Question 2:

According to Figure 1, since the variance is not constant, log-transformation can be applied to stabilize it.

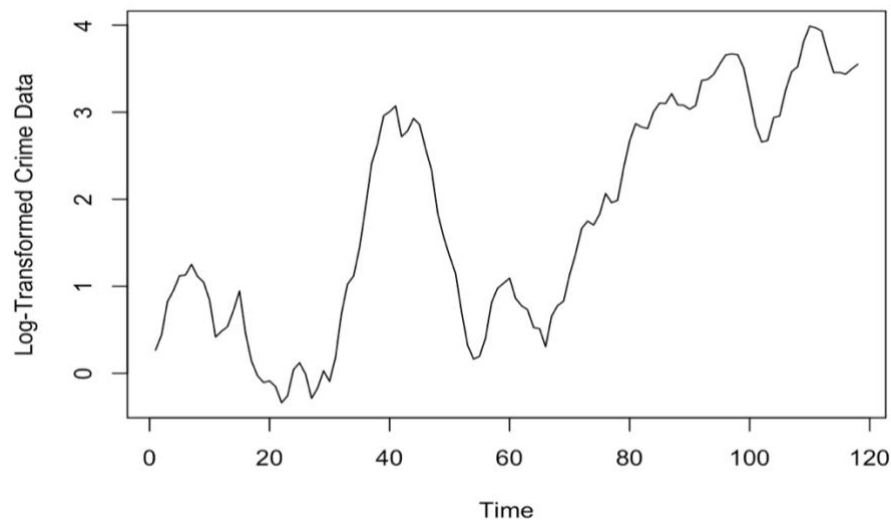


Figure 2: Log-Transformed Crime Rates Data

Based on Figure 2, the log-transformation has given the data more stability compared to the original data presented in Figure 1. The data still has an unstable mean and an upward trend present. In order to improve the stationarity by removing the trend, first order differencing can be applied.

Question 3:

Figure 2 has shown that the data is not stationary, however to confirm the results, an Augmented Dickey-Fuller (ADF) test - which tests for data stationarity - could be performed on the log-transformed crime rate data given the following results:

H_0 : the time series is not stationary (unit root is present)

Augmented Dickey-Fuller Test

```
data: crime_rates$log_rates
Dickey-Fuller = -3.0894, Lag order = 4, p-value = 0.1242
alternative hypothesis: stationary
```

According to the results of the ADF test, since the p-value is $0.1242 > 0.05$, it is therefore significant. Hence we fail to reject the H_0 . According to Figure 2 and the ADF test, we are 95% confident that the data is not stationary.

In order to improve the stationarity of the data, first order differencing can be applied to the log-transformed crime rate data to remove the upward trend still present in Figure 2.

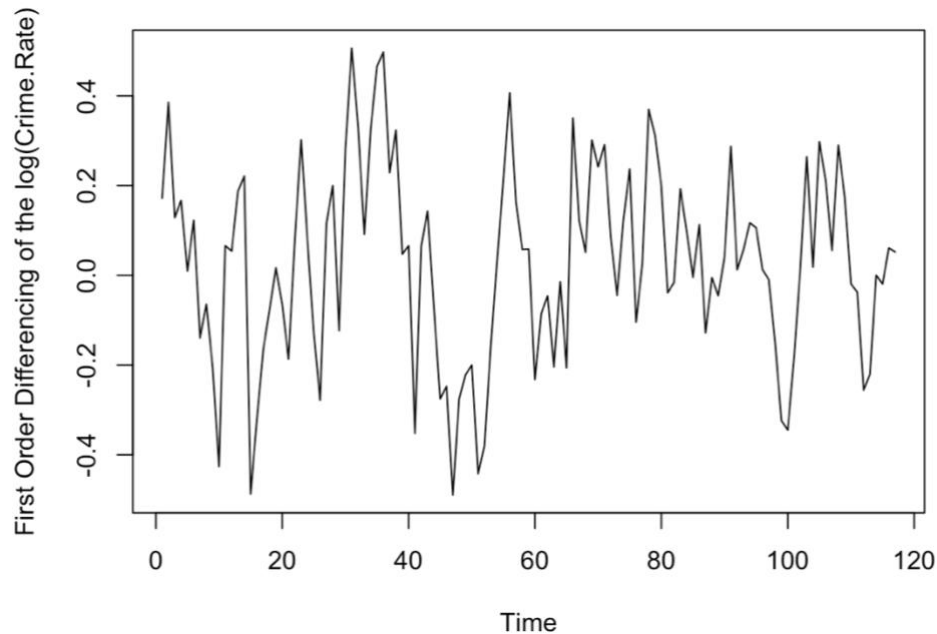


Figure 3: First Order Differenced Crime Rate Data

According to Figure 3, the mean is now centered around 0 and variance appears to be constant throughout the timeline, therefore the first order differencing has stabilized the data.

The ADF test can be performed on the differenced log-transformed crime rates data to determine whether the data is weakly stationary, resulting in the following:

Augmented Dickey-Fuller Test

```
data: first_diff
Dickey-Fuller = -3.6363, Lag order = 4, p-value = 0.03304
alternative hypothesis: stationary
```

Since the p-value is $0.033 < 0.05$, therefore we reject the H_0 . Hence at 95% confidence, the data appears to be weakly stationary.

Question 4:

ACF, also known as the Autocorrelation Function, computes the correlation between series at time t and time h . In other words it calculates the correlation between X_t and X_{t-h} including the effect of the values in between.

However, the Partial Autocorrelation Function (PACF), computes the direct correlation between X_t and X_{t-h} , excluding the effect of the values in between.

First Order Differencing of the Log-Transformed Crime Rate Data

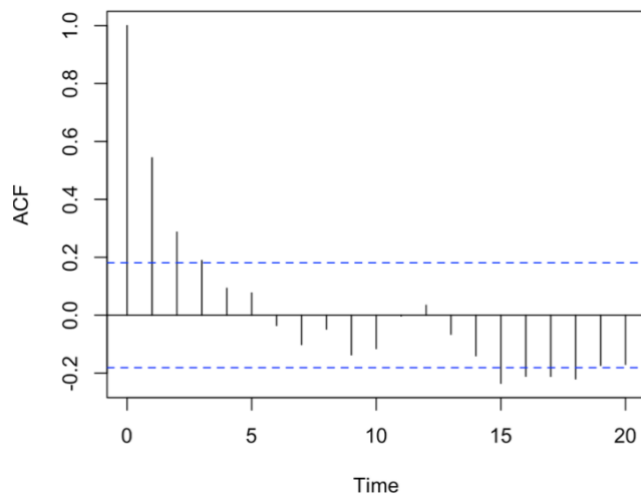


Figure 5: ACF Plot of the Log-Transformed Crime Rate Data

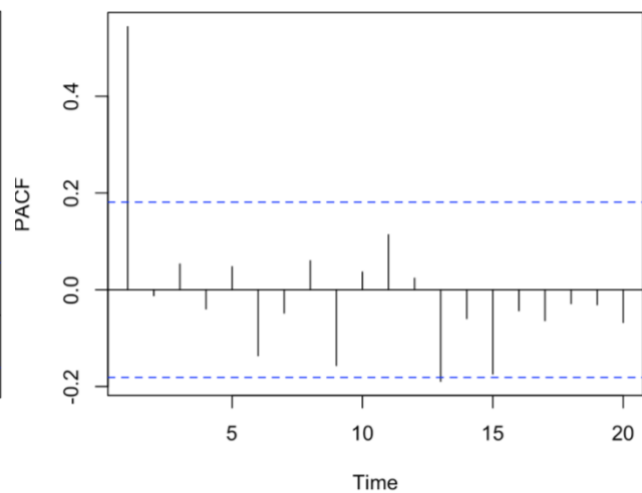


Figure 4: PACF Plot of the Log-Transformed Crime Rate Data

Question 5:

The order of an AR process, can be determined based on the PACF. According to Figure 5, the PACF plot shows a significant value at lag 0 with a cutoff at lag ,1 which indicates insignificant values thereafter and an AR(1) process.

The order of a Moving Average (MA) process could be determined based on the ACF plot. According to Figure 4, it appears that lags 1-3 are significant since they are located outside of the approximate confidence interval $[-0.2, 0.2]$. However, the ACF plot has a gradual decline which most likely indicates that any correlation after lag 1 is coming from the fact that X_t is highly dependent on X_{t-1} . Thus, the MA process does not contain much information about X_t and will not be used to avoid the model overfit.

Therefore, the proposed model for the log-transformed crime data would be ARIMA(1,1,0).

Question 6:

The proposed ARIMA(1,1,0) model assuming a non-zero mean would give the following output:

```
Series: crime_rates$log_rates
ARIMA(1,1,0) with drift

Coefficients:
      ar1    drift
      0.5414  0.0297
s.e.    0.0770  0.0362

sigma^2 = 0.03343: log likelihood = 33.62
AIC=-61.25  AICc=-61.03  BIC=-52.96
```

The ARIMA(1,1,0) model gives an AIC value of -61.25, meaning that any other model with a greater AIC value would not be a good fit to the log-transformed crime rate data.

Question 7:

Let X_t be the Gotham crime rate data and let B be the backshift operator. Writing the fitted ARIMA model in the form of:

$$(1 - \phi_1 B - \dots - \phi_p B^p)[(1 - B)^d \log(X_t) - \mu] = (1 + \theta_1 B + \dots + \theta_q B^q)\varepsilon_t$$

Where $\varepsilon_t \sim WN(0, \sigma^2)$ and μ is the estimated mean of $(1 - B)^d \log(X_t)$

$$X_t = \phi_1 X_{t-1} + \varepsilon_t \rightarrow \text{original } X_t$$

$$\log(X_t) = \phi_1 \log(X_{t-1}) + \varepsilon_t \rightarrow \text{log-transformation}$$

$$\Delta \log(X_t) = \phi_1 \Delta \log(X_{t-1}) + \varepsilon_t \rightarrow \text{first order differencing}$$

$$\Rightarrow \log(X_t) - \log(X_{t-1}) = \phi_1 [\log(X_{t-1}) - \log(X_{t-2})] + \varepsilon_t$$

$$\text{Let } Y_t = \log(X_t) - \log(X_{t-1}) = \log(X_t)(1 - B), \text{ hence } Y_{t-1} = \log(X_{t-1}) - \log(X_{t-2}) = \log(X_{t-1})B(1 - B)$$

$\therefore$

$$Y_t = \phi_1 Y_{t-1} + \varepsilon_t$$

$$\Rightarrow Y_t - \mu = \phi_1 (Y_{t-1} - \mu) + \varepsilon_t$$

$$\Rightarrow Y_t - \mu - \phi_1 (Y_{t-1} - \mu) = \varepsilon_t$$

$$\Rightarrow (Y_t - \mu) - \phi_1 B(Y_t - \mu) = \varepsilon_t$$

$$\Rightarrow (1 - \phi_1 B)(Y_t - \mu) = \varepsilon_t$$

$$\Rightarrow (1 - \phi_1 B)[(1 - B)\log(X_t) - \mu] = \varepsilon_t$$

$$\Rightarrow (1 - \phi_1 B)[(1 - B)\log(X_t) - \mu] = \varepsilon_t$$

$\therefore$ Based on the output of the proposed ARIMA(1,1,0) model, $\phi_1 = 0.5414$ and $\mu = 0.0297 \approx 0.030$, we have the following fitted model:

$$(1 - 0.541B)[(1 - B)\log(X_t) - 0.030] = \varepsilon_t$$

Question 8:

The model ARIMA(2,1,1) for the log-transformed crime rates data proposed by Mr. Bruce Wayne, has the following output:

```
Series: crime_rates$log_rates
ARIMA(2,1,1) with drift

Coefficients:
          ar1      ar2      ma1    drift
      -0.1125  0.3240  0.6786  0.0293
s.e.      0.3698  0.2253  0.3503  0.0353

sigma^2 = 0.03391:  log likelihood = 33.81
AIC=-57.61   AICc=-57.07   BIC=-43.8
```

Comparing this output with the AIC and BIC values of the proposed model ARIMA(1,1,0), it is evident that the previously proposed model is a better fit, since both AIC and BIC values are smaller compared to the Mr. Bruce Wayne's model.